

PROBIOTICS

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“Good bacteria for better health”

Probiotics are live microorganisms which, when administered in adequate amounts, confer a health benefit on the host.

Probiotics are

Live beneficial microorganisms

Mainly bacteria and some yeasts

Naturally present in the gut

Help maintain healthy intestinal flora

PROBIOTICS

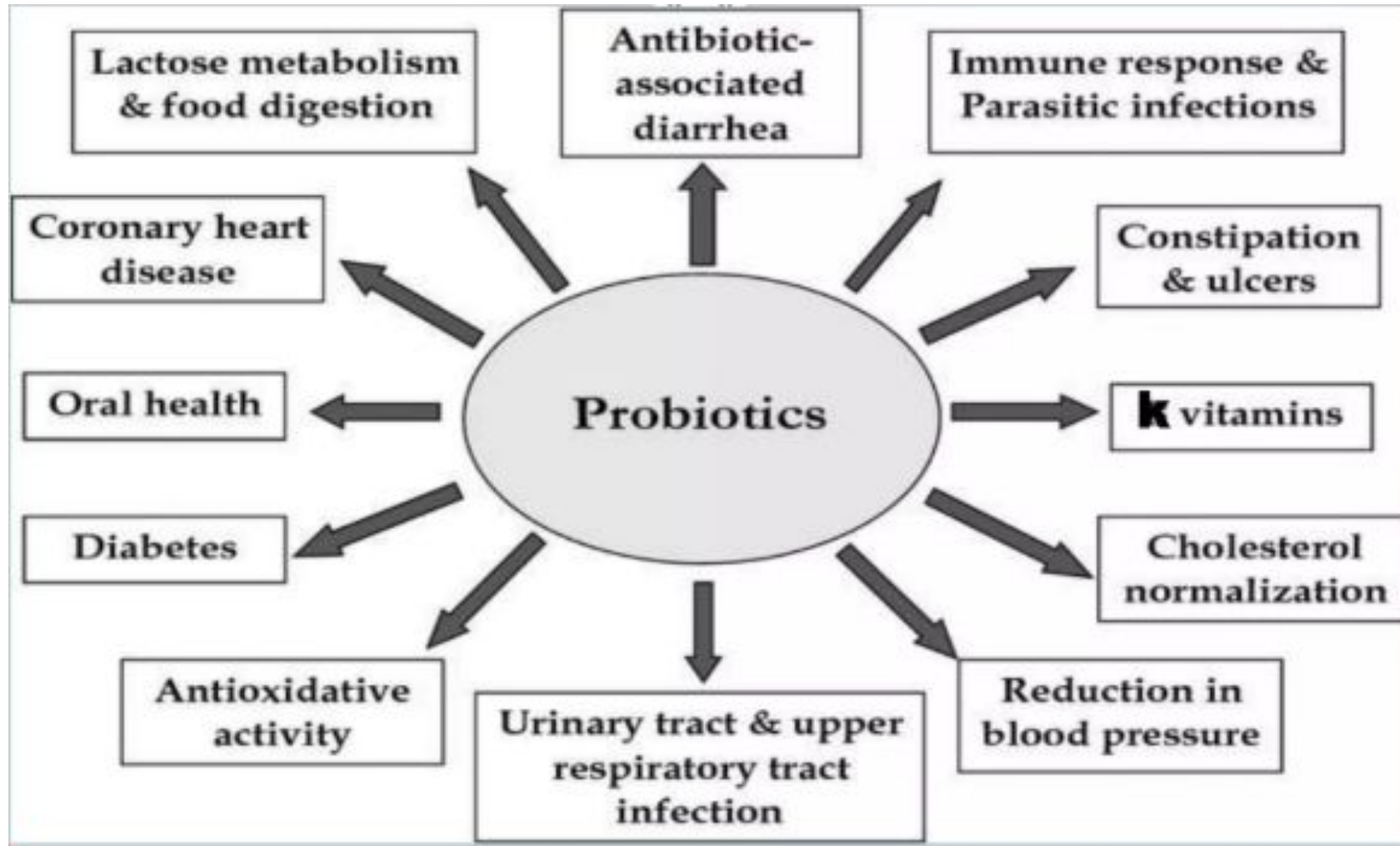
Common Probiotic Microorganisms

- *Lactobacillus* species
- *Bifidobacterium* species
- *Saccharomyces boulardii* (yeast)
- *Streptococcus thermophilus*

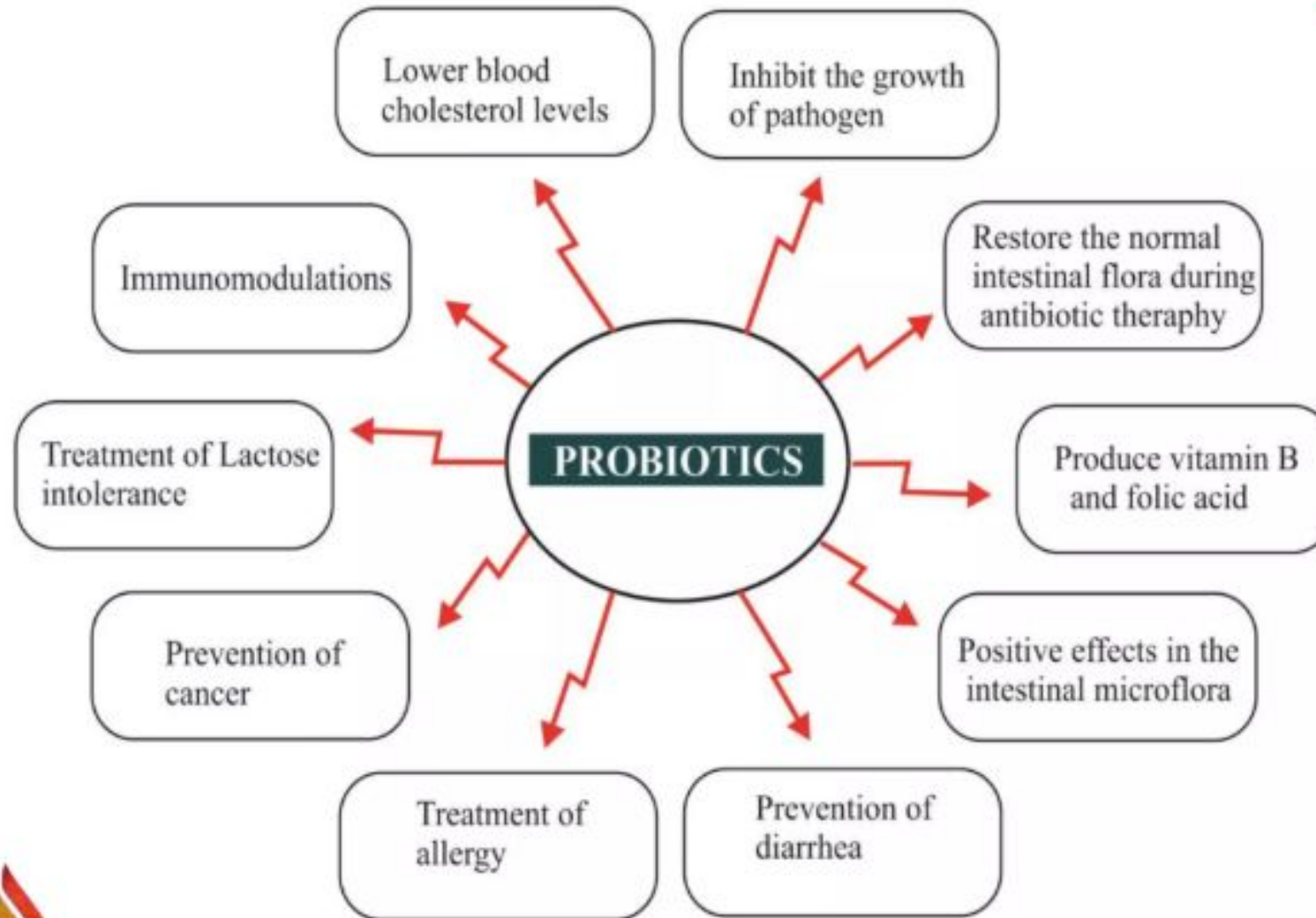
Sources of Probiotics

- Curd (yogurt)
- Buttermilk
- Kefir (fermented milk drink)
- Fermented foods (idli, dosa batter)
- Probiotic supplements

BENEFITS OF PROBIOTICS



BENEFITS OF PROBIOTICS



SIDE EFFECTS OF PROBIOTICS

- Mild gas and bloating
- Diarrhea
- Headaches
- Allergic Reactions
- Abdominal Pain
- Nausea

PREBIOTICS

Prebiotics are substance that induce the growth of microorganisms that contribute to the well-being of their host.

- A person's gut microbiome consists of trillions of living organisms called gut microbiota. The organisms are necessary for various important bodily functions, immune system support and nutrient metabolism.
- Prebiotics help to promote and sustain a healthy gut microbiome. They are non-digestible nutrients that the bacteria in the gut, including probiotics, can break down and use.
- The microbiota ferment these nutrients, which release short-chain fatty acids (SCFAs) as a by-product. SCFAs are involved in various bodily functions, including inflammation and immune system regulation.

BENEFITS OF PREBIOTICS

The benefits of prebiotics are wide-ranging.

- Help regulate bowel movements.
- Produce neurotransmitters that go back and forth between your gut and your brain to trigger mood changes and other processes.
- Stimulate your body to make hormones that aid in appetite, appetite suppression and more.
- Help your bones mineralize and absorb calcium and phosphorus, which can improve bone density.
- Improve how well your immune system functions.
- Enhance your body's anti-inflammatory response.
- Increase production of good bacteria and decrease bad bacteria that causes disease.

SOURCES OF PREBIOTICS

- **Bananas**
 - Bananas, especially when slightly unripe, are an excellent source of **resistant starch**. Resistant starch is a prebiotic fiber. This starch helps in the growth of beneficial gut bacteria and aids in digestion.
- **Garlic**
 - Garlic contains **inulin**, a fiber that acts as a prebiotic. Inulin is known for promoting gut health by enhancing the growth of friendly bacteria.
- **Onions**
 - Onions are rich in **inulin** and **fructooligosaccharides (FOS)**, These are known to promote the growth of healthy gut bacteria.
- **Chickpeas (Chana)**
 - Chickpeas are rich in **resistant starch** and **fiber**, making them a great prebiotic food. They support the growth of good bacteria and contribute to better digestion.
- **Sweet Potatoes**
 - Sweet potatoes are a rich source of **resistant starch** and **fiber**, which nourish gut bacteria.
- **Curd (Yogurt)**
 - Curd is not only a source of **probiotics** (live bacteria) but also contains **prebiotic fiber**, which nourishes these beneficial bacteria.

SOURCES OF PREBIOTICS

- **Apples**
- Apples contain **pectin**, a type of soluble fiber that acts as a prebiotic, supporting the growth of beneficial gut bacteria.
- **Jowar (Sorghum)**
- Jowar is a whole grain rich in fiber and resistant starch, which acts as a prebiotic and helps promote gut health.
- **Ragi (Finger Millet)**
- Ragi is high in **fiber** and **polyphenols**, which contribute to a healthy gut microbiome.
- **Amla (Indian Gooseberry)**
- Amla is packed with vitamin C and fiber, which help maintain gut health. It is also known for promoting digestion and detoxifying the body.
- **Bitter Gourd (Karela)**
- Bitter gourd is a good source of fiber and antioxidants that contribute to gut health by promoting the growth of beneficial bacteria.
- **Coconut**
- Both coconut meat and coconut oil contain fibers like **lauric acid**, which helps promote the growth of gut bacteria.
- **Flaxseeds**
- Flaxseeds are rich in **fiber** and **omega-3 fatty acids**, which help improve gut health by nourishing beneficial gut bacteria.

SYNBIOTICS

- The term “synbiotics” refers to any supplement that contains a blend of both prebiotics and probiotics.
- Synbiotics can help support gut health by increasing the amount of beneficial bacteria in the digestive tract while also providing these bacterial cells with the fuel they need to function and thrive.

Benefits of Synbiotics

- Support Digestive Health
- Enhance Immune Function
- Improve Mental Health
- Boost Weight Loss
- Promote Heart Health
- Decrease Inflammation

SYNBIOTICS

Risks and Side Effects

- Prebiotics are fermented in the gut and can cause adverse digestive symptoms when consumed in high amounts.
- Probiotics may also contribute to negative side effects, such as diarrhea or abdominal pain, especially when supplementation is first initiated. For this reason, it's best to start with a low amount of synbiotics and increase dosage gradually to assess tolerance.
- You should also be sure to increase your water intake when consuming more prebiotic foods. This is because prebiotic foods are typically high in fiber, which absorbs water in the colon.
- Not only can this slow down digestion, but it can also cause side effects like dehydration and constipation.
- Drinking more water can help prevent these adverse side effects and keep things running smoothly within the digestive tract.

NATURAL SYNBIOTIC FOOD COMBINATIONS

- **Yogurt + Banana**

- Yogurt → Probiotic
- Banana → Prebiotic (inulin, resistant starch)

- **Curd Rice + Onion**

- Curd → Probiotic
- Onion → Prebiotic

- **Buttermilk + Garlic**

- **Fermented Idli/Dosa + Sambar (with onions & legumes)**

- Fermented batter → Probiotics
- Legumes/onions → Prebiotics

NUTRIENT Vs NON-NUTRIENT

Aspect	Nutrient	Non-Nutrient
Definition	Chemical substances in food required for growth, energy, repair and regulation of body functions	Substances in food that are not essential nutrients but may influence health
Essentiality	Essential for survival and normal body function	Not essential for survival
Energy value	Some provide energy (carbohydrates, fats, proteins)	Usually do not provide energy
Role in body	Build tissues, provide energy, regulate metabolism	May aid digestion, protect against disease, add color/flavor
Examples	Carbohydrates, Proteins, Fats, Vitamins, Minerals, Water	Dietary fiber, Phytochemicals, Flavonoids, Tannins, Artificial colors, Preservatives

OVERNUTRITION

- Overnutrition refers to an excessive intake of dietary energy, largely from macronutrients like carbohydrates, fats, and proteins, beyond what the body expends. When energy intake consistently exceeds expenditure, the body stores this surplus.

Eg.,

- Carbohydrates are first stored as glycogen in the liver and muscles; once full, excess carbohydrates convert to fat.
- Fats, an energy-efficient fuel source, are stored as triglycerides in fat cells.
- Proteins, primarily used for tissue repair, can also convert to fat if consumed in excess.

FACTORS CONTRIBUTING TO OVERNUTRITION

- **A high intake of energy-dense foods**, particularly ultra-processed options and sugary beverages, contributes significantly to caloric excess.
- **Sedentary lifestyles** further exacerbate the problem by reducing energy expenditure. Increased reliance on modern conveniences and extended screen time means fewer opportunities for physical activity.
- **Environmental influences** also shape food choices. Easy access to unhealthy foods, often promoted through extensive marketing, can encourage overconsumption.
- **Socioeconomic factors** significantly influence dietary patterns. Individuals with lower economic status often face limitations in accessing nutritious food, relying instead on cheaper, energy-dense foods.
- **Genetic predispositions** can influence an individual's metabolism and appetite, making some more susceptible to weight gain. While genetics play a part, environmental and lifestyle factors generally exert a more substantial influence on overnutrition development.

HEALTH IMPLICATIONS OF OVERNUTRITION

Chronic overnutrition can lead to a range of serious health problems.

- **Metabolic disorders** are commonly associated with excessive nutrient intake. These include Type 2 diabetes, where cells become resistant to insulin.
- Overnutrition contributes to **high blood pressure and unhealthy cholesterol levels**, increasing the risk of heart disease and stroke.
- Overnutrition and obesity are linked to an increased risk of certain **cancers**.
- Musculoskeletal issues also arise from increased strain on joints.
- **Sleep apnea**, where breathing repeatedly stops during sleep. Excess fat deposits in the neck can narrow the airway, and increased abdominal girth can compress the lungs, making breathing difficult.
- Overnutrition can also affect **mental health**, contributing to body image concerns, negative self-esteem, and an increased risk of depression and anxiety.

CHRONIC DEGENERATIVE DISEASES

A degenerative disease is a condition characterized by the progressive and irreversible deterioration of cells, tissues, and organs. This degradation occurs over time and can result in loss of function, disability, or cell death.

Degenerative diseases can be classified according to the affected tissue or organ, such as :

- **Neurodegenerative Diseases:** A group of complex diseases characterized by neuronal loss and the progressive degeneration of different areas of the nervous system.
- **Musculoskeletal Degenerative Diseases:** Diseases that generally cause degeneration of the joints, cartilage, and bone structures, leading to pain, inflammation, stiffness, or loss of mobility.
- **Cardiovascular Degenerative Diseases:** Conditions in which there is a progressive accumulation of atheromatous plaques in the arteries, leading to reduced blood flow and increasing the risk of events such as heart attacks and strokes.

MAIN DEGENERATIVE DISEASES

Alzheimer's Disease (AD)

- It is the most common form of **neurodegenerative disease** and the leading cause of dementia, representing 60-80% of cases. This condition is characterized by neuronal loss and cognitive dysfunction and is rapidly becoming one of the most costly, deadly, and burdensome diseases of the century.

Parkinson's Disease (PD)

- It is a chronic and progressive neurodegenerative disorder that primarily affects older adults, although it can also appear in younger patients. It is the second most common neurodegenerative disease, affecting about 4 million people worldwide.

Osteoporosis

- Osteoporosis is characterized by low bone mass and microstructural deterioration of bone tissue, which increases bone fragility and raises the risk of fractures.

Multiple Sclerosis (MS)

- In **multiple sclerosis**, the immune system attacks the myelin, the protective layer around nerves in the brain and spinal cord due to inflammation. This damage to the myelin interferes with communication between the brain and the rest of the body, causing delayed or blocked nerve signals, which can lead to a variety of neurological symptoms and impact quality of life and functional capacity.